

Experiment No: 2

Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required:

JavaScript, HTML, CSS and Code Editor.

Theory:

JavaScript Variables and Modern ES6 Features:

Variables are used in JavaScript to store and manage data values. JavaScript provides three ways to declare variables: var, let, and const. Modern JavaScript (ES6 and later) also introduces features such as template literals and destructuring, which make code more readable and efficient.

1. var Keyword

The var keyword is the traditional way of declaring variables in JavaScript. Variables declared with var are function-scoped and can be redeclared and updated within their scope.

2. let Keyword

The let keyword is used to declare block-scoped variables. It allows updating the variable value but does not allow redeclaration within the same scope.

3. const Keyword

The const keyword is used to declare constants whose values cannot be reassigned after initialization. It is also block-scoped.

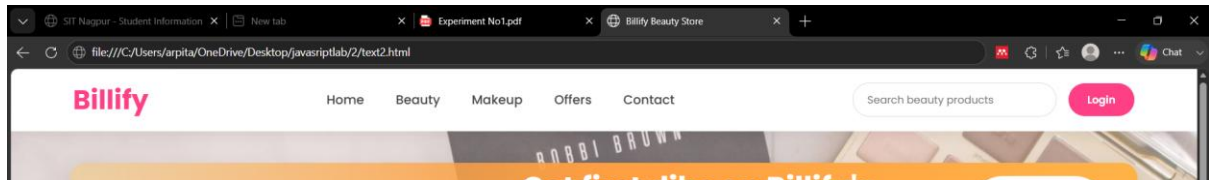
Template Literals:

Template literals are a modern JavaScript feature that allows embedding variables and expressions directly within strings using backticks (```) and `${}` syntax. They improve readability and simplify string concatenation.

Destructuring

Destructuring is a convenient way to extract values from arrays or properties from objects and assign them to variables.

Output



Case Study Program Code

Problem Statement

Design a **Beauty Store Billing Calculator** that accepts customer details, product information, quantity, discount, coupon code, and payment method. The system should calculate the subtotal, apply discounts and GST, and display the final bill dynamically.

Implementation

The webpage consists of an HTML form where the user enters billing details. JavaScript is used to retrieve the entered data using **DOM methods**, perform bill calculations, apply discounts, coupon offers, and GST, and generate the invoice dynamically using **template literals** and **innerHTML**. Modern JavaScript features such as **var**, **let**, **const**, and **object destructuring** are used to make the code more efficient and readable. The **Generate Bill** button calls the JavaScript function through an **inline event (onclick)**, while the billing logic is written in an **external JavaScript file**.

Conclusion:

JavaScript variables (**var**, **let**, and **const**) provide flexible ways to store and manage data. Modern features such as template literals and destructuring simplify coding and improve readability. By using these features, a billing calculator can efficiently process user input, perform calculations, and generate accurate billing information dynamically.